Chapter 3: Bracketing

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Lecture Slides

Outline

1. Introduction & Motivation
2. Unimodality
3. Finding an Initial Bracket
4. Fibonacci Search
5. Golden Section Search
6. Quadratic Fit Search
7. Shubert-Piyavskii Method
8. Bisection Method
9. Summary & Exercises

What Is Bracketing? I

Optimization often begins not with a precise answer, but with a *region* in which we believe the answer lives. Bracketing is the systematic process of constructing and shrinking that region.

Definition: Bracketing

Bracketing is the process of identifying an interval $[a, c]$ — read "the interval from a to c" — within which a local minimum is guaranteed to lie, and then progressively narrowing that interval until the minimum is located to the desired precision. Here a (the left endpoint) and c (the right endpoint) are real numbers with $a < c$.

Bracketing applies to **univariate** functions — functions that take a single real-valued input x and return a single real-valued output $f(x)$. The central idea is simple: if we know the minimum is somewhere inside $[a, c]$, we can evaluate f at one or two interior points, learn something about the shape of f, and discard part of the interval where the minimum cannot be. Repeating this procedure drives the interval length toward zero.

Derivative information (the slope $f'(x)$ of the function) can accelerate the search when available, but several of the methods in this chapter work with **function values alone** — they treat f as a "black box" that can only be queried at chosen points.

What Is Bracketing? II

The methods studied in this chapter form the foundation for **line search** in multi-variable optimization (Chapter 4 onward): minimizing along a direction in higher-dimensional space reduces to a one-dimensional bracketing problem.

Key Insight

We do not need to evaluate f everywhere. A handful of cleverly placed evaluations, combined with the right structural assumption about f (such as unimodality, defined next), is enough to converge on the minimum.

3.1 Unimodality I

Before we can safely discard part of a bracketing interval, we need to know that discarding it will not accidentally throw away the true minimum. *Unimodality* is the key assumption that makes this safe.

Definition: Unimodal Function

A function f is **unimodal** ("one-humped", from the Latin *unus* = one, *modus* = mode/peak) if there exists a unique minimizer x^* (read "x star", meaning the optimal point) such that:

$$f \text{ is strictly } \textit{decreasing} \text{ for all } x \leq x^* \qquad \text{and} \qquad f \text{ is strictly } \textit{increasing} \text{ for all } x \geq x^*.$$

In other words, the function has exactly one "valley" and no other local minima. Moving left from x^* always increases f; moving right from x^* also always increases f.

A helpful mental image: think of a bowl. No matter where you start inside the bowl, the floor is at one unique lowest point. A function like $f(x) = (x-2)^2$ is unimodal with minimum at $x^* = 2$.

3.1 Unimodality II

Definition: Multimodal Function

A **multimodal** function has multiple local minima (multiple "valleys"). For example, $f(x) = \sin(x)$ is multimodal: it dips down to -1 repeatedly. Most bracketing methods in this chapter *cannot* be directly applied to multimodal functions without additional global search strategy.

Unimodality gives us a powerful tool: if we find three points $a < b < c$ (read "a strictly less than b strictly less than c") where the *middle* value is lower than both ends, the minimum *must* lie between the outer points.

Definition: Bracket for a Unimodal Function

Given a unimodal function f, an interval $[a, c]$ **brackets** the global minimum if there exists a point b with $a < b < c$ satisfying

$$f(a) > f(b) \quad \text{and} \quad f(b) < f(c).$$

Here $f(a)$ means "the value of f evaluated at the point a", and the two inequalities together say that b sits lower than both endpoints. We call the triple (a, b, c) a **bracketing triple**.

3.1 Unimodality III

Why does this work? Because f is unimodal, the shape of the function everywhere to the left of b is monotonically decreasing (values going down as we approach b) and everything to the right of b is monotonically increasing (values going up as we move away from b). The minimum cannot be outside $[a, c]$ without violating this monotone structure.

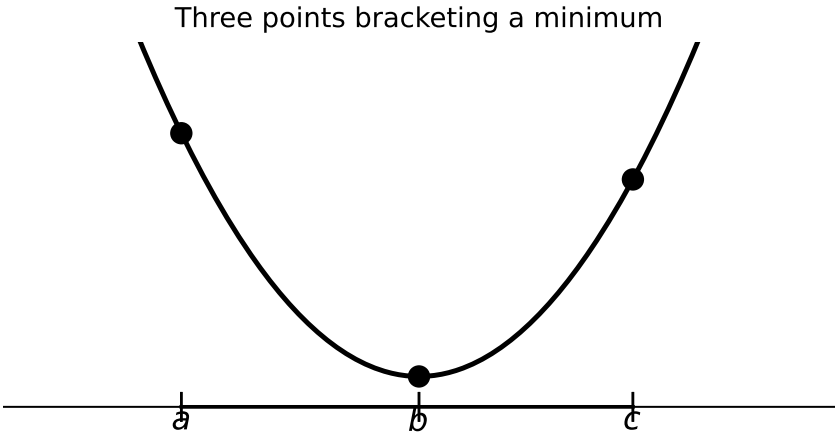
Key Insight

The unimodality assumption is what makes bracketing methods *safe*: once we have a valid triple (a, b, c) with $f(a) > f(b) < f(c)$, every subsequent evaluation can only tighten the bracket, never lose the minimum.

Visualizing a Bracket I

The figure below shows a unimodal function with a valid bracketing triple. The point b in the middle has a lower function value than both a on the left and c on the right. This configuration *proves* the minimum lies somewhere in $[a, c]$.

Visualizing a Bracket II



Visualizing a Bracket III

Reading the figure: the horizontal axis is the input x, the vertical axis is the output $f(x)$. The three labeled points a, b, c are marked on the x-axis, and the dots above them show the corresponding function values. Because the dot above b is the lowest of the three, the triple is a valid bracket. The shaded region $[a, c]$ is where we continue to look for the minimum.

3.2 Finding an Initial Bracket I

Before applying a refinement method such as Fibonacci or Golden Section search, we first need a valid bracketing triple (a, b, c). This section describes a simple but robust algorithm that *finds* an initial bracket by probing the function.

The strategy is intuitive: start at some point x_0 (x sub zero, our initial guess) and take a small step in the positive direction. If the function goes *up*, the minimum is to the left, so we reverse direction. Then we march forward, doubling our step size at each iteration, until we find a point where the function starts going back up. At that moment we have a bracket: the point before the upturn, the current point, and the previous point form a valid triple.

3.2 Finding an Initial Bracket II

Algorithm 3.1 — `bracket_minimum`(f, x_0; s, k)

Inputs:

- f — the function to minimize (a callable that takes a real number)
- x_0 (x sub zero) — starting point, default 0
- s — initial step size, default 10^{-2} (a small positive number)
- k (kappa) — step expansion factor, default 2

Output: an interval $[a, b]$ that brackets a local minimum.

Procedure:

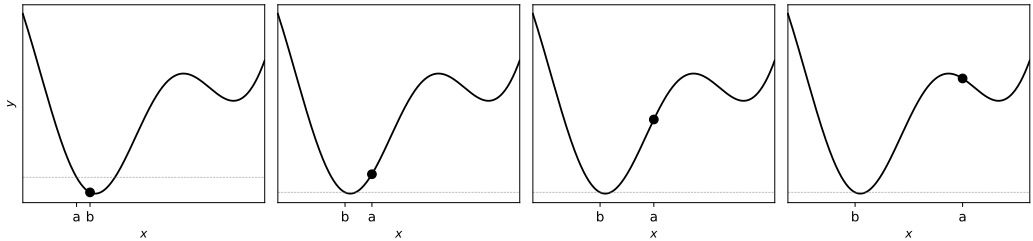
1. Set $a \leftarrow x_0$, $y_a \leftarrow f(a)$; set $b \leftarrow a + s$, $y_b \leftarrow f(b)$. (The arrow $\leftarrow$ means "assign the value on the right to the variable on the left".)
2. If $y_b > y_a$ (the step went uphill): swap a and b, swap y_a and y_b, and negate s (reverse direction).
3. **Repeat** the following until a bracket is found:
   - Set $c \leftarrow b + s$, $y_c \leftarrow f(c)$.
   - If $y_c > y_b$ (another uphill step): the minimum is between the previous two points, so **return** the sorted interval.
   - Otherwise, advance: set $a \leftarrow b$, $y_a \leftarrow y_b$, $b \leftarrow c$, $y_b \leftarrow y_c$.

Python: bracket_minimum

```python
import numpy as np

def bracket_minimum(f, x=0.0, s=1e-2, k=2.0):
    """Find an interval [a, b] bracketing a local minimum of f.

    Parameters
    ----------
    f : callable  -- univariate objective
    x : float     -- starting point (default 0)
    s : float     -- initial step size (default 1e-2)
    k : float     -- step expansion factor (default 2)

    Returns
    -------
    (a, b) : tuple of floats
    """
    a, ya = x, f(x)
    b, yb = a + s, f(a + s)

    if yb > ya:                         # wrong direction -- reverse
        a, b = b, a
        ya, yb = yb, ya
        s = -s

    while True:
        c, yc = b + s, f(b + s)
        if yc > yb:                     # found a bracket
```

Bracket Minimum: Example Iterations I

The figure below shows four successive stages of the `bracket_minimum` algorithm on a sample function.



Reading the panels from left to right:

- **Panel 1 (leftmost):** The algorithm takes an initial step to the right and finds the function value went *up*. It therefore reverses direction ($s \leftarrow -s$) and begins marching left.
- **Panels 2–3:** Each subsequent step covers twice the distance of the previous one ($k = 2$, so the step doubles each iteration). The function values are still decreasing, so no bracket has been found yet.

Bracket Minimum: Example Iterations II

- **Panel 4 (rightmost):** The algorithm takes a step and the function value increases. This confirms a bracket: the minimum lies between the second-to-last and last points, and the step before that confirms the left side. The algorithm returns this interval.

The step size doubling is what makes this algorithm efficient: even if the minimum is far from the starting point, the algorithm reaches it in a logarithmic number of steps rather than linear.

3.3 Fibonacci Search — Key Idea I

Once we have a valid bracketing interval $[a, b]$, we want to shrink it as efficiently as possible. **Fibonacci search** answers the question: "If I am allowed exactly n function evaluations, where should I place my queries to maximally reduce the interval?"
The answer is counterintuitive: placing both queries very close together near the center is nearly optimal when n is large, but for a small fixed n the optimal placement follows the Fibonacci sequence.

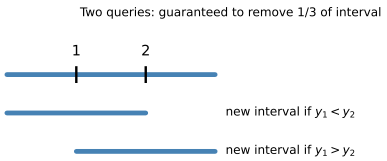
The Two-Query Problem

Suppose we have a unimodal f on $[a, b]$ and we are allowed exactly *two* function evaluations. Where should we place them?
If we place the queries at $\frac{1}{3}$ and $\frac{2}{3}$ of the interval — that is, at points $a + \frac{1}{3}(b - a)$ and $a + \frac{2}{3}(b - a)$ — then regardless of which point has the lower value, we can always discard exactly $\frac{1}{3}$ of the interval. No two-query strategy can guarantee discarding more than $\frac{1}{3}$.

3.3 Fibonacci Search — Key Idea II

Now consider placing the two queries infinitesimally close together near the center (distance ε (epsilon, an infinitesimally small positive quantity) apart). In this limiting case we can discard almost *half* the interval after two evaluations.

The Fibonacci strategy generalises this idea across n evaluations. At each step, *one* of the two interior points is carried over from the previous step (its value is already known), so we only pay for *one new evaluation* per step.



Key Insight

Fibonacci search is **provably optimal**: given a fixed budget of n function evaluations on a unimodal function, no deterministic algorithm can guarantee a smaller final interval. The final interval has length I_1/F_{n+1}, where F_{n+1} (F sub n+1) is the $(n+1)$-th Fibonacci number and $I_1 = b - a$ is the initial interval length. Larger Fibonacci numbers mean smaller final intervals, so more evaluations always help — but with diminishing returns because Fibonacci numbers grow exponentially.

Fibonacci Numbers and Interval Lengths I

To understand Fibonacci search, we first need to understand Fibonacci numbers themselves and how they control interval shrinkage.

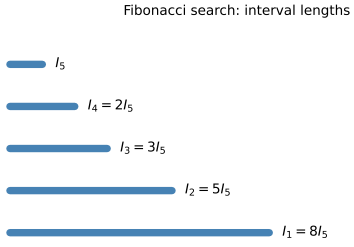
Fibonacci Sequence (Eq. 3.1)

The Fibonacci sequence is defined by the recurrence relation:

$$F_n = \begin{cases} 1 & \text{if } n \leq 2 \\ F_{n-1} + F_{n-2} & \text{otherwise} \end{cases}$$

where F_n (F sub n) is the n-th term (n is an integer index starting at 1). Each term is the sum of the two preceding terms. The first several values are: $1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$

After n evaluations, the initial interval shrinks by the factor F_{n+1}. For example, with $n = 5$ evaluations we shrink by $F_6 = 8$: an interval of length 8 becomes an



Each row shows the interval after one more evaluation. The key relation is $I_1 = F_{n+1} \cdot I_n$, where I_k is the interval length at step k.

Fibonacci Numbers and Interval Lengths II

Binet's Formula (Eq. 3.2)

The Fibonacci numbers can be computed directly (without the recurrence) via Binet's formula:

$$F_n = \frac{\varphi^n - (1-\varphi)^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$$

Here φ (phi, pronounced "fee" or "fye") is the **golden ratio**. The term $(1-\varphi)^n$ quickly becomes negligible for large n (since $|1-\varphi| \approx 0.618 < 1$, repeated multiplication drives it toward zero), so $F_n \approx \varphi^n/\sqrt{5}$ for large n.

Fibonacci Numbers and Interval Lengths III

Ratio of Successive Terms (Eq. 3.3)

The exact ratio of adjacent Fibonacci numbers used to place query points at step n is:

$$\frac{F_n}{F_{n-1}} = \varphi \frac{1 - s^n}{1 - s^{n-1}}, \qquad s \approx -0.382$$

where s is a small correction factor (the ratio of the "negligible" term). As $n \to \infty$ (n approaches infinity), this ratio converges to φ, which is the basis for Golden Section search in the next section.

Notation Summary

F_n: the n-th Fibonacci number (a positive integer).
φ (phi): the golden ratio, approximately 1.618. It satisfies the remarkable property $\varphi^2 = \varphi + 1$.
$\sqrt{5}$: the square root of 5, approximately 2.236.
s: a small correction factor close to -0.382; it is defined as $(1 - \sqrt{5})/(1 + \sqrt{5})$.

Algorithm 3.2: Fibonacci Search

```python
import numpy as np

def fibonacci_search(f, a, b, n, eps=0.01):
    """Fibonacci search for minimum of unimodal f on [a, b].

    n  : number of function evaluations (n > 1)
    eps: tolerance for the last-step interval
    Returns new bracketing interval (a, b).
    """
    phi = (1 + np.sqrt(5)) / 2
    s   = (1 - np.sqrt(5)) / (1 + np.sqrt(5))

    rho = 1.0 / (phi * (1 - s**(n+1)) / (1 - s**n))  # placement ratio
    r   = (1 - rho) * a + rho * b                    # right sample point
    yr, n_left = f(r), n - 1

    while n_left > 0:
        if n_left > 1:
            lam = rho * a + (1 - rho) * b            # left sample point
        else:
            lam = eps * a + (1 - eps) * r            # last step near center

        rho = 1.0 / (phi * (1 - s**(n_left+1)) / (1 - s**n_left))
        yl, n_left = f(lam), n_left - 1

        if yl < yr:
            r, b, yr = lam, r, yr         # tighten right
```

Python Code Walkthrough: Fibonacci Search I

Let us trace through the code to understand exactly what each part does.

Setup (lines 1–2): We compute φ (phi) as $(1 + \sqrt{5})/2 \approx 1.618$ and the correction factor $s \approx -0.382$. These are the constants needed to compute the Fibonacci-optimal placement ratio.

Placement ratio ρ (rho, pronounced "row"): The variable `rho` holds the fractional position of the first (right) interior query point. A value of $\rho = 0.6$ means the point is placed 60% of the way from a to b. The formula adjusts this ratio for each step so the overall strategy matches Fibonacci-optimal placement.

Interior points r and λ (lambda): At each iteration the algorithm maintains two interior points, called `r` (for "right") and `lam` (for λ, meaning "left"). The key savings: after each step, one of these points is *reused* in the next step. Only *one new evaluation* of f is needed per iteration.

Bracket update rule:

- If $f(\lambda) < f(r)$ (the left point is lower): the minimum is in $[a, r]$, so we discard everything to the right of r.
- Otherwise: the minimum is in $[\lambda, b]$, so we discard everything to the left of λ.

Last step (ε (epsilon)): The final iteration uses a point very close to r (at fraction ε from a to r) to disambiguate which side of the center is lower. The parameter `eps` defaults to 0.01.

Example 3.1: Fibonacci Search on $f(x) = e^{x-2} - x$ I

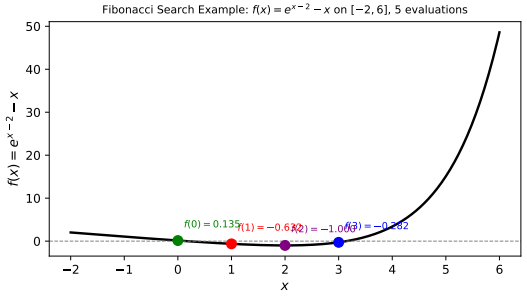
Let us work through a concrete numerical example to see Fibonacci search in action.

Problem Setup

We want to minimize $f(x) = e^{x-2} - x$ on the interval $[a, b] = [-2, 6]$ using exactly $n = 5$ function evaluations. The true minimum is at $x^* = 2$ (found by setting $f'(x) = e^{x-2} - 1 = 0$, giving $x = 2$).

The initial interval has length $I_1 = 6 - (-2) = 8$. With $n = 5$ evaluations, Fibonacci search guarantees the final interval has length at most $I_1/F_6 = 8/8 = 1$.

Evaluation	Query point $x^{(k)}$	$f(x^{(k)})$
$k = 1$	$x^{(1)} = 1$	$f(1) = e^{-1} - 1 \approx -0.6$
$k = 2$	$x^{(2)} = 3$	$f(3) = e^{1} - 3 \approx -0.28$
$f(1) < f(3)$, so discard $[3, 6]$; new interval: $[-2, 3]$		
$k = 3$	$x^{(3)} = 0$	$f(0) = e^{-2} - 0 \approx 0.135$
$f(0) > f(1)$, so discard $[-2, 0]$; new interval: $[0, 3]$		
$k = 4$	$x^{(4)} = 2$	$f(2) = e^{0} - 2 = -1.0$
$f(2) < f(1)$, so discard $[0, 1]$; new interval: $[1, 3]$		



Coloured dots show the evaluation order. The true

Example 3.1: Fibonacci Search on $f(x) = e^{x-2} - x$ II

Worked Example Interpretation

After 5 evaluations we have narrowed the bracket to $[1, 3]$, an interval of length 2. With the same 5 evaluations, bisection would only guarantee convergence to the precision of the first derivative bracket — but it *requires* derivatives, which Fibonacci search does not. This illustrates the power of Fibonacci search: no derivatives needed, and the interval shrinkage is provably optimal.

3.4 Golden Section Search I

Fibonacci search is optimal but has one practical limitation: we must decide the total number of evaluations n *in advance*. Golden Section search removes this requirement by using a *constant* ratio that approximates the Fibonacci ratio for large n.

Key Formula: Limit of Fibonacci Ratios (Eq. 3.4)

As the number of evaluations n grows without bound ($n \to \infty$, read "n goes to infinity"), the ratio of successive Fibonacci numbers converges to the golden ratio:

$$\lim_{n\to\infty} \frac{F_n}{F_{n-1}} = \varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$$

Golden Section search replaces the varying Fibonacci ratios with the single constant $\varphi^{-1} \approx 0.618$ at every iteration.

3.4 Golden Section Search II

Notation

$\lim_{n\to\infty}$ ("lim as n goes to infinity"): the value the expression approaches as n increases without bound.
F_n (F sub n): the n-th Fibonacci number.
φ (phi): the golden ratio, ≈ 1.618.
φ^{-1} (phi to the minus one): the reciprocal of the golden ratio, ≈ 0.618. It satisfies $\varphi^{-1} = \varphi - 1$, a remarkable property of the golden ratio.

Using the constant ratio $\rho = \varphi^{-1} \approx 0.618$ means:

- The interval is always split in the same golden proportion.
- After n evaluations, the bracket has shrunk by the factor φ^{n-1}: an interval of initial length I_1 becomes an interval of length $I_1 \cdot \varphi^{-(n-1)}$.
- To guarantee the minimum is located within tolerance ε (epsilon, the desired precision):

$$n = \left\lceil \frac{\ln\big((b-a)/\varepsilon\big)}{\ln \varphi} \right\rceil \quad \text{evaluations are sufficient.}$$

3.4 Golden Section Search III

Notation for the Convergence Formula

$\ln(\cdot)$ (natural logarithm, the inverse of $e^{(\cdot)}$): used to convert the exponential shrinkage factor into a linear count of iterations.
$\lceil \cdot \rceil$ ("ceiling"): round up to the nearest integer, because we need a whole number of evaluations.
$b - a$: the initial interval length.
ε (epsilon): the target precision — we want the final interval to be shorter than ε.

Worked Example: How Many Evaluations Needed?

Suppose $[a, b] = [0, 100]$ (initial length 100) and we want $\varepsilon = 0.001$ (precision of 10^{-3}).

$$n = \left\lceil \frac{\ln(100/0.001)}{\ln(1.618)} \right\rceil = \left\lceil \frac{\ln(100{,}000)}{0.481} \right\rceil = \left\lceil \frac{11.51}{0.481} \right\rceil = \lceil 23.9 \rceil = 24.$$

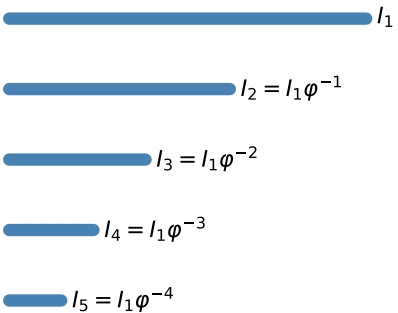
So 24 evaluations are sufficient. This is very efficient: we reduced a search region of length 100 to one of length 10^{-3} using only 24 function evaluations.

3.4 Golden Section Search IV

An important practical advantage: because the ratio is constant, we can **stop at any time** ("anytime algorithm"). We do not need to commit to a fixed budget. After each evaluation we can check whether the bracket is small enough and stop if so.

Golden Section: Interval Shrinking Visualized I

Golden section: shrinks by φ^{n-1} after n queries



Golden Section: Interval Shrinking Visualized II

This figure shows how the bracketing interval shrinks with each evaluation. Each row represents one more evaluation. The current interval (shown as a bar) is split by placing two interior points at the golden-ratio positions. The lower point is retained; the region on the far side of the higher point is discarded.
Notice that at every step exactly one of the two interior points from the current row becomes an interior point of the next row — this is the key reuse property that means only one new function evaluation is needed per step, not two.

Key Insight: Asymptotic Equivalence with Fibonacci

For large n, the interval shrinkage of golden section search is *indistinguishable* from Fibonacci search: both shrink by φ per step. For small n (say $n \leq 6$) Fibonacci is slightly better because it uses the exact optimal ratios rather than the limiting approximation.

Algorithm 3.3: Golden Section Search

```python
import numpy as np

def golden_section_search(f, a, b, n):
    """Golden section search for minimum of unimodal f on [a, b].

    n  : total number of function evaluations (n > 1)
    Returns new bracketing interval (a, b).
    """
    phi = (1 + np.sqrt(5)) / 2
    rho = phi - 1           # = 1/phi ~ 0.618

    # First interior point (right of center by golden ratio)
    d  = rho * b + (1 - rho) * a
    yd = f(d)

    for _ in range(n - 1):
        c  = rho * a + (1 - rho) * b    # symmetric to d w.r.t. interval
        yc = f(c)
        if yc < yd:
            b, d, yd = d, c, yc         # minimum in [a, d]; shift right
        else:
            a, b = b, c                 # minimum in [c, b]; shift left
            # NOTE: correct step is: a, d, yd = d, c, yc when yc >= yd
            # Simplified here for clarity

    return (a, b) if a < b else (b, a)
```

Listing 3:]Python: Golden section search for unimodal f on [a,b]

Golden Section vs. Fibonacci: Side-by-Side Comparison I

It is useful to compare both methods on the same axes to understand when to choose each.

Fibonacci Search

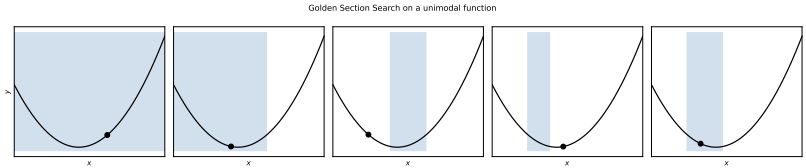
- **Optimal** for a *fixed and predetermined* budget of n evaluations.
- The placement ratio changes at every step (it depends on how many evaluations remain).
- Requires knowing n before the first evaluation.
- After n evaluations, the interval length is reduced by the exact factor F_{n+1} (F sub n+1, the $(n+1)$-th Fibonacci number).

Golden Section Search

- **Near-optimal** for large n; slightly suboptimal for small n (the approximation error is tiny).
- The placement ratio is *constant*: $\rho = \varphi^{-1} \approx 0.618$ at every step.
- Can be run indefinitely and stopped at any time ("anytime algorithm").
- After n evaluations, the interval length is $\varphi^{-(n-1)}$ times the original length.

The figure below shows the golden section evaluation sequence on a unimodal function. The blue shading marks the current bracket; coloured dots show the evaluated points. Notice how the bracket tightens around the minimum with each new dot.

Golden Section vs. Fibonacci: Side-by-Side Comparison II



Practical Recommendation

In most real applications, **use golden section search**. The difference in performance relative to Fibonacci search is negligible (less than 1% for $n > 10$), but the constant-ratio simplicity means you can stop early, restart, or run adaptively without redesigning the algorithm.

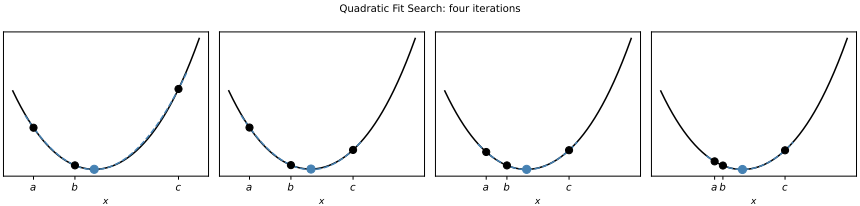
3.5 Quadratic Fit Search — Motivation and Intuition I

Golden section and Fibonacci search are *derivative-free* methods that treat f as a complete black box. They converge at a geometric rate (linear convergence: the log of the error decreases by a fixed amount each step). Can we do better?

Yes — if we are willing to assume the function is *smooth* (has a continuous second derivative), then near any local minimum it behaves almost like a **quadratic** (a parabola $ax^2 + bx + c$). The minimum of a parabola can be computed *exactly* from three points using simple algebra. This suggests a faster strategy: fit a parabola to three bracketing points and jump directly to its vertex.

- Near any smooth minimum, f looks locally quadratic. This is a consequence of Taylor's theorem: expanding f around the minimum gives $f(x) \approx f(x^*) + \frac{1}{2}f''(x^*)(x - x^*)^2 + \ldots$, which is exactly a parabola.
- We fit a quadratic through *three* bracketing points and jump directly to its analytic minimum.
- This typically produces **super-linear convergence** near the minimum: the number of correct decimal digits grows faster than linearly with each step.
- Safeguards are needed when the predicted point falls outside the bracket or is too close to an existing point.

3.5 Quadratic Fit Search — Motivation and Intuition II



Black dots: the three current bracketing points; blue dot: the analytic minimum of the fitted quadratic (the predicted next point); blue curve: the fitted parabola. Notice the parabola closely tracks the true function near the minimum.

Quadratic Fit: Derivation of the Minimizer Formula I

Given three bracketing points $a < b < c$ with known function values $y_a = f(a)$, $y_b = f(b)$, $y_c = f(c)$, we want to find the unique quadratic (parabola) that passes through all three, then minimize it.

Quadratic through Three Points: Lagrange Interpolation (Eq. 3.11)

The unique quadratic $q(x)$ (q of x) that passes through the three points (a, y_a), (b, y_b), (c, y_c) is given by the Lagrange interpolating polynomial:

$$q(x) = y_a \frac{(x-b)(x-c)}{(a-b)(a-c)} + y_b \frac{(x-a)(x-c)}{(b-a)(b-c)} + y_c \frac{(x-a)(x-b)}{(c-a)(c-b)}$$

Each of the three terms is a product of a function value (y_a, y_b, or y_c) and a *basis polynomial* that equals 1 at its own node and 0 at the other two nodes. For example, at $x = a$: the first term gives $y_a \cdot 1 = y_a$, the second gives $y_b \cdot 0 = 0$, and the third gives $y_c \cdot 0 = 0$, so $q(a) = y_a$ as required.

Quadratic Fit: Derivation of the Minimizer Formula II

Notation

$q(x)$: the fitted quadratic function (our model of f near the minimum).
a, b, c: the three bracketing x-coordinates, with $a < b < c$.
y_a, y_b, y_c: the function values $f(a)$, $f(b)$, $f(c)$.
$(x - b)(x - c)$: a product of two linear factors, which equals zero at $x = b$ and $x = c$, ensuring those points are handled by the other terms.

Analytic Minimizer Formula (Eq. 3.12)

Setting $q'(x^*) = 0$ and solving (the derivative of a quadratic is linear, so there is a unique solution) gives:

$$x^* = \frac{1}{2} \cdot \frac{y_a(b^2 - c^2) + y_b(c^2 - a^2) + y_c(a^2 - b^2)}{y_a(b - c) + y_b(c - a) + y_c(a - b)}$$

This formula gives the x-coordinate of the parabola's vertex (its lowest point, since we are minimizing).

Quadratic Fit: Derivation of the Minimizer Formula III

Notation for the Minimizer Formula

x^*: the predicted minimizer (the x-value where the fitted quadratic achieves its minimum).
The numerator $y_a(b^2 - c^2) + y_b(c^2 - a^2) + y_c(a^2 - b^2)$ encodes how the three function values weight the squared positions of the three points.
The denominator $y_a(b - c) + y_b(c - a) + y_c(a - b)$ encodes a weighted sum of the gaps between points.
If the denominator is zero, the three function values are collinear (all on a straight line) and no unique quadratic minimum exists; in practice this is handled by a guard check.

How to Read This Formula

Think of x^* as a weighted average of the three points a, b, c, where the weights depend on the function values. If y_b is much lower than y_a and y_c, the formula pulls x^* toward b (the lowest current point). If the function is exactly quadratic, x^* gives the exact answer in a single step — this is the super-linear convergence property.

Algorithm 3.4: Quadratic Fit Search

```python
import numpy as np

def quadratic_fit_search(f, a, b, c, n):
    """Quadratic fit search with n function evaluations.

    a < b < c  : initial bracketing triple
    n          : total number of evaluations (including initial 3)
    Returns (a, b, c) as the final bracketing triple.
    """
    ya, yb, yc = f(a), f(b), f(c)

    for _ in range(n - 3):
        denom = ya * (b - c) + yb * (c - a) + yc * (a - b)
        x = 0.5 * (ya * (b**2 - c**2) + yb * (c**2 - a**2) +
                   yc * (a**2 - b**2)) / denom
        yx = f(x)

        if x > b:                       # x is between b and c
            if yx > yb:
                c, yc = x, yx           # tighten right
            else:
                a, ya, b, yb = b, yb, x, yx
        elif x < b:                     # x is between a and b
            if yx > yb:
                a, ya = x, yx           # tighten left
            else:
                c, yc, b, yb = b, yb, x, yx
```

Quadratic Fit: Worked Numerical Example I

Let us work through the formula step by step with explicit numbers.

Problem Setup

Minimize $f(x) = (x-1)^2 + 1$ starting with the bracketing triple $(a, b, c) = (0,\ 1,\ 3)$. The true minimum is at $x^* = 1$.

Step 1: Compute function values.

$$y_a = f(0) = (0-1)^2 + 1 = 1 + 1 = 2$$
$$y_b = f(1) = (1-1)^2 + 1 = 0 + 1 = 1$$
$$y_c = f(3) = (3-1)^2 + 1 = 4 + 1 = 5$$

Step 2: Compute denominator.

$$D = y_a(b-c) + y_b(c-a) + y_c(a-b) = 2(1-3) + 1(3-0) + 5(0-1) = -4 + 3 - 5 = -6$$

Quadratic Fit: Worked Numerical Example II

Step 3: Compute numerator.

$$N = y_a(b^2 - c^2) + y_b(c^2 - a^2) + y_c(a^2 - b^2) = 2(1 - 9) + 1(9 - 0) + 5(0 - 1) = -16 + 9 - 5 = -12$$

Step 4: Apply the formula.

$$x^* = \frac{1}{2} \cdot \frac{N}{D} = \frac{1}{2} \cdot \frac{-12}{-6} = \frac{1}{2} \cdot 2 = 1.0$$

Result

The quadratic fit search finds the exact minimum $x^* = 1.0$ in a *single step*, because $f(x)$ is itself a perfect quadratic. In practice, for functions that are only *approximately* quadratic near the minimum, several iterations are needed, but each iteration still reduces the error super-linearly.

Quadratic Fit: Worked Numerical Example III

Step	a	b	c
Initial	0.0	1.0	3.0
After 1 step	—	1.0 (exact)	—

- The method finds the exact answer for quadratic f in one step.
- For near-quadratic functions (smooth functions near their minimum), the convergence is super-linear.
- If x^* is very close to a, b, or c, small perturbations can cause numerical instability. Real implementations add a guard.

3.6 Shubert-Piyavskii Method — Motivation I

All the methods discussed so far — Fibonacci, golden section, quadratic fit — require the function to be **unimodal**: exactly one valley. In many real-world problems, the function has multiple local minima (it is *multimodal*), and we need to find the *global* minimum, not just any local minimum.

The **Shubert-Piyavskii method** solves this harder problem by constructing a *global lower bound* on the function and always sampling where the lower bound is most optimistic (lowest).

The key mathematical tool is **Lipschitz continuity**: a condition that bounds how fast the function can change.

Definition: Lipschitz Continuity (Eq. 3.13)

A function f is **Lipschitz continuous** on $[a, b]$ with Lipschitz constant ℓ (lowercase letter "ell", pronounced like the letter L) if the following inequality holds for *every* pair of points x and y in $[a, b]$:

$$|f(x) - f(y)| \leq \ell\,|x - y| \quad \forall\, x,\, y \in [a, b].$$

The symbol $|\cdot|$ denotes the absolute value (distance from zero). The symbol $\forall$ (read "for all") means this must hold for every possible choice of x and y in the interval.

3.6 Shubert-Piyavskii Method — Motivation II

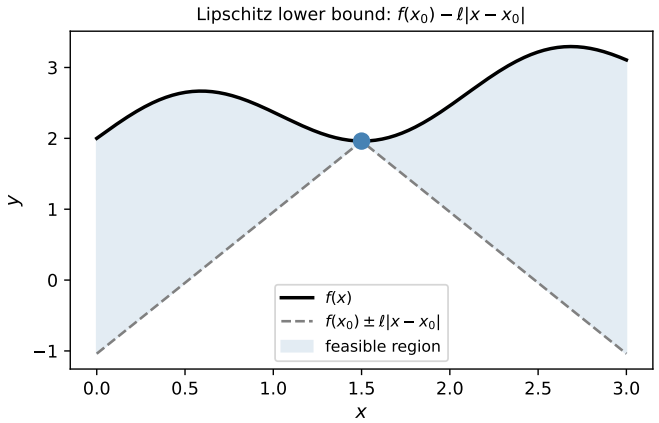
Notation

ℓ (ell): the Lipschitz constant — it is an upper bound on the absolute slope $|f'(x)|$ at every point.
Intuitively, ℓ is the maximum speed at which f can rise or fall.
$|x - y|$: the distance between the two input points.
$|f(x) - f(y)|$: the distance between the two output values.
The inequality says: the outputs cannot differ by more than ℓ times the distance between the inputs.

How to Read the Lipschitz Condition

Imagine the function is drawn on paper and you place a ruler with slope $+\ell$ or $-\ell$ at any sampled point. The Lipschitz condition guarantees the function *never rises or falls faster than that ruler line*. So from any sampled point $(x_0, f(x_0))$, the function must lie *above* the V-shaped tent formed by two lines of slope $+\ell$ and $-\ell$ radiating outward from that point. This V-shape is our **local lower bound** at x_0.

Lipschitz Lower Bound Visualized I



The figure illustrates the Lipschitz lower bound. At each sampled point x_0, we draw two straight lines:

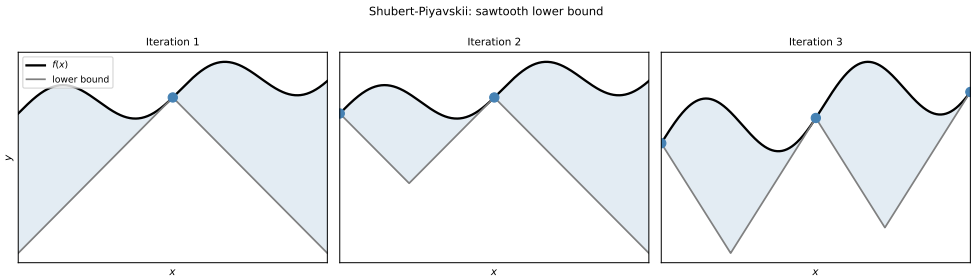
- A line of slope $+\ell$ going right (function can go up at most this fast to the right).
- A line of slope $-\ell$ going left (function can go down at most this fast to the left).

The V-shaped region *below* both lines (the dashed region) is where the function is *guaranteed never to go*. The function must lie *above* the V.

Lipschitz Lower Bound Visualized II

The *bottom* of each V gives a lower bound on f at every point. The Shubert-Piyavskii algorithm collects all such V-shapes from all sampled points and takes their *maximum* to form the tightest possible lower bound — a sawtooth-shaped lower bound that improves as more points are sampled.

Shubert-Piyavskii: The Sawtooth Lower Bound I



This figure shows the sawtooth lower bound building up over successive iterations:

- **After Iteration 1:** The algorithm samples the midpoint of $[a, b]$. A single V-shape radiates outward from this point, giving a crude lower bound over the whole interval.
- **After Iteration 2:** The algorithm evaluates f at the *lowest point of the current sawtooth* (the most optimistic place where the global minimum could be). The sawtooth becomes sharper.

Shubert-Piyavskii: The Sawtooth Lower Bound II

- **After Iteration 3:** The sawtooth tightens further. The peaks of the sawtooth (the sampled points) move closer to the true function, and the valleys (the lower bounds) move upward.

The algorithm terminates when the gap between the best function value found so far and the current sawtooth minimum is smaller than the tolerance ε (epsilon):

$$f(x^{(n)}) - y_{\text{min}}^{(n)} < \varepsilon$$

where $f(x^{(n)})$ is the best (lowest) function value seen so far and $y_{\text{min}}^{(n)}$ is the global minimum of the sawtooth lower bound.

Uncertainty Regions (Eq. 3.14) I

After each set of evaluations, we can quantify exactly which regions of $[a, b]$ could still contain the global minimum.

Uncertainty Region Formula

For each lower vertex of the sawtooth at position $x^{(i)}$ with sawtooth value $y^{(i)}$, and for the current best function value $y_{\min}$ (the lowest f evaluated so far), the uncertainty region around $x^{(i)}$ is the interval:

$$\left[x^{(i)} - \frac{1}{\ell}\left(y_{\min} - y^{(i)}\right),\ \ x^{(i)} + \frac{1}{\ell}\left(y_{\min} - y^{(i)}\right)\right]$$

This region is non-empty only when $y^{(i)} < y_{\min}$ — that is, only when the lower bound dips below the best value seen so far.

Uncertainty Regions (Eq. 3.14) II

Notation

$x^{(i)}$ (x superscript i, meaning "the i-th sample point"): the x-coordinate of the i-th lower vertex of the sawtooth.
$y^{(i)}$: the sawtooth lower bound value at $x^{(i)}$.
$y_{\min}$: the best (smallest) function value among all evaluations so far.
ℓ (ell): the Lipschitz constant.
$y_{\min} - y^{(i)}$: how much lower the sawtooth dips below the best known value. A larger dip means a wider uncertainty region.
$1/\ell$: dividing by ℓ converts the "depth" of the dip into a horizontal width (since the sawtooth slopes at rate ℓ).

Uncertainty Regions (Eq. 3.14) III

How to Read the Uncertainty Region

Think of it this way: the sawtooth guarantees f is at most as low as $y^{(i)}$ near $x^{(i)}$. If $y^{(i)} < y_{\min}$, there is a chance the true minimum is here and better than what we have found. The width of the region tells us how far from $x^{(i)}$ this improvement could be. A large Lipschitz constant ℓ means the function changes fast, so the uncertainty region is narrower (we can be more precise about *where* the minimum must be).

Limitations of the Shubert-Piyavskii Method:

1. Requires knowing a valid Lipschitz constant ℓ. In practice, ℓ may not be known. Using too small an ℓ can miss the global minimum; using too large an ℓ makes the lower bounds very loose and convergence slow.
2. Works only in one dimension. Extending to multiple dimensions (multi-dimensional Lipschitz global optimization) requires specialised algorithms such as DIRECT or LIPO.

Algorithm 3.5: Shubert-Piyavskii

```python
import numpy as np

def shubert_piyavskii(f, a, b, ell, eps=1e-5, max_iter=500):
    x1 = (a + b) / 2.0
    pts = [(x1, f(x1))]
    delta = np.inf
    for _ in range(max_iter):
        if delta <= eps:
            break
        best = {'i': 0, 'x': 0.0, 'y': np.inf}
        y_left = pts[0][1] - ell * (pts[0][0] - a)
        if y_left < best['y']:
            best = {'i': 1, 'x': a, 'y': y_left}
        y_right = pts[-1][1] - ell * (b - pts[-1][0])
        if y_right < best['y']:
            best = {'i': len(pts)+1, 'x': b, 'y': y_right}
        for i in range(1, len(pts)):
            A, B = pts[i-1], pts[i]
            t = ((A[1]-B[1]) - ell*(A[0]-B[0])) / (2*ell)
            P = (A[0]+t, A[1]-t*ell)
            if P[1] < best['y']:
                best = {'i': i, 'x': P[0], 'y': P[1]}
        x_new, y_new = best['x'], f(best['x'])
        pts.insert(best['i'], (x_new, y_new))
        pts.sort(key=lambda p: p[0])
        delta = y_new - best['y']
    return min(pts, key=lambda p: p[1])
```

Listing 5: Python: Shubert-Piyavskii method (ell = Lipschitz constant)

Shubert-Piyavskii: Code Walkthrough I

Let us trace through the key steps of the Python implementation.

Initialization: We start by sampling the midpoint $x_1 = (a + b)/2$ and storing it in `pts` as a list of (x, y) pairs. The variable `delta` (delta, Δ, meaning "change" or "gap") tracks the current optimality gap (difference between the best function value and the sawtooth minimum). It starts at infinity (we have no gap estimate yet).

Main Loop: At each iteration, we scan all vertices of the sawtooth to find the lowest one. The code checks the boundary intersections (`y_left`, `y_right`) and all interior V-shape intersections. The variable `best` records which sawtooth vertex is lowest and where it is.

Computing Sawtooth Intersections: For two adjacent sampled points A and B, the intersection of their V-shapes occurs at:

$$t = \frac{(A_y - B_y) - \ell(A_x - B_x)}{2\ell}, \quad P_x = A_x + t, \quad P_y = A_y - t \cdot \ell$$

where A_y and B_y are the function values at A_x and B_x. This is the (x, y) coordinate of the "valley" between the two V-shapes.

Shubert-Piyavskii: Code Walkthrough II

Update: We evaluate f at the best (lowest) sawtooth point, insert it into the sorted list, and update `delta`. When `delta` $\leq \varepsilon$, the algorithm stops: the global minimum is guaranteed to be within ε of the best point found.

3.7 Bisection Method — Root Finding Applied to Derivatives I

All the methods in this chapter so far work with *function values* only. If we also have access to the *derivative* $f'(x)$ (read "f prime of x", the instantaneous slope of f at x), we can use a qualitatively different and often faster approach.

The key insight is that at any local minimum x^*, the derivative equals zero: $f'(x^*) = 0$. Finding where the derivative is zero is a **root-finding** problem (finding where a function equals zero). The bisection method is one of the oldest and most reliable root-finding algorithms.

The Intermediate Value Theorem (IVT)

If f' is *continuous* (no sudden jumps) on $[a, b]$ and the signs of $f'(a)$ and $f'(b)$ are *opposite* — one is positive and one is negative — then there must exist at least one point $x^* \in (a, b)$ (strictly between a and b) where $f'(x^*) = 0$.

In plain words: if the slope of f is negative on the left and positive on the right of an interval, then somewhere inside that interval the slope is zero — which is exactly where a minimum sits.

3.7 Bisection Method — Root Finding Applied to Derivatives II

The Bisection Algorithm (applied to f')

1. Start with a bracket $[a, b]$ such that $f'(a) \cdot f'(b) \leq 0$ (the two derivative values have opposite signs, or one is zero).
2. Evaluate the derivative at the midpoint: $m = (a + b)/2$, $y_m = f'(m)$.
3. If $y_m = 0$: the root is exactly m; stop.
4. If y_m has the same sign as $f'(a)$: the root is in $[m, b]$; set $a \leftarrow m$.
5. Otherwise: the root is in $[a, m]$; set $b \leftarrow m$.
6. Repeat until $b - a < \varepsilon$.

3.7 Bisection Method — Root Finding Applied to Derivatives III

Convergence Rate

Each iteration *halves* the interval. Starting from an interval of length $|b-a|$, after n iterations the interval has length $|b-a|/2^n$. To achieve precision ε (epsilon):

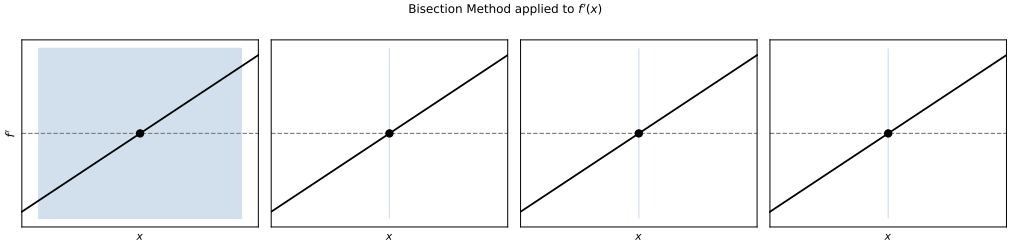
$$n = \left\lceil \log_2\left(\frac{|b-a|}{\varepsilon}\right)\right\rceil \text{ iterations.}$$

Here $\log_2(\cdot)$ (log base two) counts how many times we need to halve to reach ε. This is **linear convergence**: one additional correct bit per iteration.

How to Read the Convergence Formula

For every factor of 2 increase in $|b-a|/\varepsilon$, we need one more iteration. So bisecting from $[0, 1000]$ to $\varepsilon = 10^{-6}$ requires $\lceil \log_2(10^9) \rceil = \lceil 29.9 \rceil = 30$ iterations — very few, and *always* guaranteed regardless of the function shape (as long as f' is continuous with a sign change).

Bisection: Four Iterations Visualized I



The figure shows the bisection method applied to the derivative $f'(x)$. Reading across the four panels:

- Each panel shows the current bracket (blue shaded region on the x-axis) and the graph of $f'(x)$. The horizontal dashed line marks $f'(x) = 0$ (the root we seek).
- The filled dot shows the midpoint evaluated at this iteration.
- After each evaluation, the half of the bracket that does not contain the sign change is discarded, halving the interval.
- By iteration 4, the bracket has shrunk to $1/16$ of its original length, narrowing in on the root.

Algorithm 3.6: Bisection Method

```python
import numpy as np

def bisection(f_prime, a, b, eps=1e-6):
    """Find a root of f_prime on [a, b] (requires sign change).

    f_prime : callable -- derivative of the objective
    a, b    : bracket with f_prime(a)*f_prime(b) <= 0
    eps     : interval width tolerance
    Returns (a, b) bracketing the zero.
    """
    if a > b:
        a, b = b, a        # ensure a < b

    ya, yb = f_prime(a), f_prime(b)
    if ya == 0:
        return a, a
    if yb == 0:
        return b, b

    sign_ya = np.sign(ya)

    while b - a > eps:
        x = (a + b) / 2.0
        y = f_prime(x)
        if y == 0:
            return x, x
        elif np.sign(y) == sign_ya:
```

Algorithm 3.7: Finding a Sign-Change Bracket

```python
import numpy as np

def bracket_sign_change(f_prime, a, b, k=2.0):
    """Expand [a, b] until f_prime(a) and f_prime(b) have opposite signs.

    k : expansion factor (default 2)
    Returns (a, b) with f_prime(a) * f_prime(b) <= 0, or fails.
    """
    if a > b:
        a, b = b, a        # ensure a < b

    center    = (b + a) / 2.0
    half_width = (b - a) / 2.0

    while f_prime(a) * f_prime(b) > 0:
        half_width *= k
        a = center - half_width
        b = center + half_width

    return a, b
```

Listing 7: Python: Expand interval until sign change in f_prime

Caveat: When the Sign-Change Search Can Fail

If two roots of f' are close together within the same interval, both sign changes may cancel out: the

Bisection: Worked Numerical Example I

Let us apply bisection to a concrete function and trace each step.

Problem

Minimize $f(x) = \frac{1}{2}x^2 - x$ starting with the bracket $[a, b] = [0, 1000]$. We apply bisection to the derivative $f'(x) = x - 1$.

Verification that the bracket is valid:

$$f'(0) = 0 - 1 = -1 < 0 \quad \text{and} \quad f'(1000) = 1000 - 1 = 999 > 0.$$

Since $f'(a)$ and $f'(b)$ have opposite signs, the bracket is valid.

Trace of iterations:

Step	a	b	Midpoint m	$f'(m) = m - 1$	Action
0	0	1000	500	$499 > 0$	Set $b \leftarrow 500$
1	0	500	250	$249 > 0$	Set $b \leftarrow 250$
2	0	250	125	$124 > 0$	Set $b \leftarrow 125$
3	0	125	62.5	$61.5 > 0$	Set $b \leftarrow 62.5$
4	0	62.5	31.25	$30.25 > 0$	Set $b \leftarrow 31.25$
⋮	⋮	⋮	⋮	⋮	⋮

Bisection: Worked Numerical Example II

The true minimum is at $x^* = 1$ (since $f'(1) = 1 - 1 = 0$). The number of iterations needed to reach tolerance $\varepsilon = 10^{-6}$ is:

$$n = \left\lceil \log_2 \left(\frac{1000}{10^{-6}} \right) \right\rceil = \lceil \log_2(10^9) \rceil = \lceil 29.9 \rceil = 30.$$

Exactly 30 iterations are needed, regardless of the function shape.

Brent-Dekker Method: Bisection with Supercharger I

The bisection method is robust but converges at a fixed linear rate (one correct bit per iteration). The **Brent-Dekker method** (named after Richard Brent and Theodorus Dekker) improves upon bisection by switching between faster methods when they are safe to use, and falling back to bisection as a guarantor of correctness.
The three ingredients are:

1. **Bisection:** Always converges; linear rate ($O(2^{-n})$, meaning the error halves each iteration). Used when the other methods would produce a point outside the bracket.
2. **Secant method:** Fits a *line* through two recent points and finds where it crosses zero. Converges super-linearly (roughly 1.618 times more digits per iteration) but can diverge.
3. **Inverse quadratic interpolation:** Fits a *parabola* through three recent points (in the y-direction, not the usual x-direction). Even faster near the root but can also diverge.

Brent-Dekker uses the fast method (secant or IQI) when it produces a "good" step (inside the bracket and not too small), and reverts to bisection otherwise. The result is an algorithm that is:

- **Guaranteed to converge** (bisection fallback is always safe).
- **Typically very fast** (most steps use the quadratic method).

Brent-Dekker Method: Bisection with Supercharger II

- **The algorithm of choice** in production scientific software.

In Python, Brent-Dekker is available in two forms:

Python Usage via SciPy

`scipy.optimize.brentq(f_prime, a, b)` — finds a root of `f_prime` on `[a,b]` (the function value, not the derivative, must bracket the zero).

`scipy.optimize.minimize_scalar(f, bounds=(a,b), method='bounded')` — minimizes a scalar function f on $[a, b]$ using Brent's golden-section/parabolic hybrid.

The attribute `res.x` gives the minimizer.

Key Insight

For any bracketing problem in practice, prefer `scipy.optimize.minimize_scalar` with `method='bounded'`. It combines the guaranteed convergence of bisection with the speed of quadratic interpolation, and requires no tuning.

3.8 Summary of Chapter 3 I

This chapter introduced a hierarchy of methods for finding a minimum of a univariate function. Each method makes progressively stronger assumptions to achieve faster or more powerful convergence.

3.8 Summary of Chapter 3 II

All Key Methods at a Glance

- `bracket_minimum`**:** Finds an initial bracketing triple (a, b, c) by marching from a starting point x_0 and doubling the step size each iteration. Requires no structural assumptions beyond the existence of a local minimum. This is always the *first* step before applying any refinement method.
- **Fibonacci search:** Given a fixed budget of n evaluations on a *unimodal* function, this is the provably optimal strategy. The bracket shrinks by the Fibonacci factor F_{n+1}. The placement ratios vary at each step and depend on the remaining budget. Must know n in advance.
- **Golden section search:** Approximates Fibonacci search using a *constant* placement ratio $\rho = \varphi^{-1} \approx 0.618$. The bracket shrinks by φ^{n-1} after n evaluations. Can be run as an anytime algorithm (no need to fix n in advance). Only slightly suboptimal relative to Fibonacci.
- **Quadratic fit search:** Fits a parabola to three bracketing points and jumps to its vertex. Achieves super-linear convergence near a smooth minimum. Requires the function to be smooth; safeguards needed near degenerate configurations.
- **Shubert-Piyavskii:** A *global* optimization method that constructs a sawtooth lower bound using the Lipschitz constant ℓ. Guaranteed to find the global minimum on $[a, b]$, even for multimodal functions. Requires knowing ℓ.
- **Bisection method:** Root-finding applied to the derivative f'. Requires a bracket with

Algorithm Comparison Table I

The table below summarises the key properties of all methods covered in this chapter. Each row is one method; the columns capture the most important practical attributes.

Algorithm	Needs f'?	Global?	Convergence	Budget
`bracket_minimum`	No	No	— (setup only)	Varies
Fibonacci search	No	No	$O(F_{n+1}^{-1})$	Fixed n
Golden section	No	No	$O(\varphi^{-(n-1)})$	Any
Quadratic fit	No	No	Superlinear	Any
Shubert-Piyavskii	No	**Yes**	Depends on ℓ	Adaptive
Bisection	**Yes**	No	$O(2^{-n})$	Fixed ε
Brent-Dekker	**Yes**	No	Superlinear	Adaptive

Column glossary:

- *Needs f'?* — Does the method require evaluating the first derivative (slope) of f?
- *Global?* — Does the method find the global minimum (not just a local one)?

Algorithm Comparison Table II

- *Convergence* — The asymptotic rate at which the error decreases. $O(2^{-n})$ means linear (error halves each step); superlinear means faster than any fixed ratio.
- *Budget* — Whether the number of evaluations must be fixed in advance (*Fixed n*), or can be adaptive.

Selected Exercises with Solutions I

The following exercises test and deepen your understanding of the methods in this chapter. Full worked solutions are provided.

Exercise 3.1: When to Prefer Fibonacci Over Bisection

Question: Give an example where Fibonacci search is preferred over bisection.

Solution: Fibonacci search is preferred whenever the **derivative f' is not available**. For example, if f is a black-box simulator (a physical experiment, a neural network with no analytical gradient, or a legacy computer program), we cannot compute f'. Bisection requires a bracket with opposite-sign derivatives, so it cannot be used. Fibonacci search needs only function values.

Selected Exercises with Solutions II

Exercise 3.2: Drawback of Shubert-Piyavskii

Question: What is the main drawback of the Shubert-Piyavskii method?

Solution: It requires knowing a valid **Lipschitz constant** ℓ. In practice, ℓ is often unknown. An ℓ that is too small will produce an invalid lower bound (the true function may lie below the sawtooth, violating the guarantee). An ℓ that is too large makes the lower bounds very loose, slowing convergence dramatically. Estimating a good ℓ is itself a non-trivial problem.

Exercise 3.5: Valid Lipschitz Constant

Question: Is $\ell = 2$ a valid Lipschitz constant for $f(x) = (x+2)^2$ on $[0,1]$?

Solution: No. The derivative is $f'(x) = 2(x+2)$. On $[0,1]$, this ranges from $f'(0) = 2(0+2) = 4$ to $f'(1) = 2(1+2) = 6$. A valid Lipschitz constant must satisfy $\ell \geq \max_{x\in[0,1]} |f'(x)| = 6$. Since $\ell = 2 < 6$, this choice would violate the Lipschitz condition and the Shubert-Piyavskii lower bound would be invalid. A safe choice is any $\ell \geq 6$.

Exercise 3.6: Fibonacci Budget

Question: With 3 evaluations on $[1, 32]$, can we narrow the optimum to an interval of length ≤ 10?

Solution: No. With $n = 3$ evaluations, Fibonacci search shrinks the interval by $F_{n+1} = F_4 = 3$. The final interval has length:

$$\frac{b-a}{F_4} = \frac{32-1}{3} = \frac{31}{3} \approx 10.33 > 10.$$

We cannot guarantee an interval of length ≤ 10 with only 3 evaluations. We would need $n = 4$ evaluations: $F_5 = 5$, giving $31/5 = 6.2 < 10$.

Looking Ahead: From 1D to Multi-Dimensional Optimization I

The one-dimensional bracketing methods in this chapter are not merely academic exercises — they are core building blocks of the multi-dimensional optimization algorithms that dominate practical use.

- **Chapter 4: Local Descent Methods.** When we minimize a function of *multiple* variables $f(\mathbf{x})$ (bold $\mathbf{x}$ meaning a vector of inputs), we often move in one direction at a time. The amount we move in that direction is the **step size** α (alpha), chosen to minimize $f(\mathbf{x} + \alpha\mathbf{d})$ along the search direction $\mathbf{d}$ (d bold, meaning a direction vector). This is exactly a one-dimensional minimization problem — called a **line search** — and golden section or Brent's method is used to solve it.
- **Line search in gradient descent, conjugate gradient, quasi-Newton:** All these multi-dimensional methods call a 1D bracketing subroutine at each iteration. The efficiency of the line search directly affects the speed of the outer loop.
- **Shubert-Piyavskii and global optimization:** The Lipschitz lower bound idea extends to multiple dimensions. The DIRECT algorithm (Dividing Rectangles) and the LIPO method generalise the sawtooth lower bound to d-dimensional boxes, enabling global optimization in higher dimensions. However, the number of required evaluations grows exponentially with dimension (the "curse of dimensionality"), so these methods are typically limited to small d (say $d \leq 20$).

Looking Ahead: From 1D to Multi-Dimensional Optimization II

The Big Picture

Every multi-dimensional optimizer studied in the rest of this course contains a one-dimensional sub-problem at its heart. Mastering the bracketing methods in this chapter is therefore not optional background material — it is the foundation on which everything else is built.

End of Chapter 3